Sachin Ashok

sachin.cs.illinois.edu sachinashok.g@gmail.com
LinkedIn Google Scholar GitHub

Research Interests

Networked Systems; with an emphasis on Microservices and Cloud Computing.

Education

2021 - **PhD.** in Computer Science
University of Illinois at Urbana-Champaign
Advisors: **Prof. Radhika Mittal**, **Prof. Philip Brighten Godfrey**

2014 - 2018 **B.Tech.** in Computer Science and Engineering
National Institute of Technology, Trichy

Research Experience

2018 - 2020 **Microsoft Research, India** **Research Fellow**
Mentors: **Venkat Padmanabhan**, **Johannes Gehrke**

- **iBox: Internet in a Box**: Designed a data-informed network simulator for recreating the behaviour of paths in a target network such as the Internet.
  Published at SIGMETRICS'22.
- **R3Net: Reinforcement Learning for Bandwidth Control**: Developed realistic network environments to train RL agents for bandwidth estimation & congestion control.
  Published at MLForSystems'19.

Publications

- **Fast and Efficient Look-Ups via Data-Driven FIB Designs**
  Sachin Ashok*, Aditi Partap*, Ammar Tahir* (* = equal contribution)
  In Proceedings of ACM SIGCOMM FIRA Workshop, 2022.
- **Data-Driven Network Path Simulation with iBox**
  Sachin Ashok, Shubham Tiwari, Nagarajan Natarajan, Venkata N. Padmanabhan, Sundararajan Sellamanickam
  In Proceedings of ACM SIGMETRICS, 2022.
- **Leveraging Service Meshes as a New Network Layer**
  Sachin Ashok, P. Brighten Godfrey, and Radhika Mittal
  In Proceedings of ACM HotNets, 2021.
- **iBox: Internet in a Box**
  Sachin Ashok, Sai Surya Duvvuri, Nagarajan Natarajan, Venkat Padmanabhan, Sundararajan Sellamanickam, Johannes Gehrke
  In Proceedings of ACM HotNets, 2020.
- **Reinforcement Learning for Bandwidth Estimation and Congestion Control in Real-Time Communications**
  Joyce Fang, Martin Ellis, Bin Li, Siyao Liu, Yasaman Hosseinkashi, Michael Revow, Albert Sadovnikov, Ziyuan Liu, Peng Cheng, **Sachin Ashok**, David Zhao, Ross Cutler, Yan Lu, Johannes Gehrke
  In Proceedings of NeurIPS MLForSystems workshop, 2019.

Industry Experience

2016 - 2017 **Mozilla Organization** **Intern**
Mentors: **Franziskus Kiefer**, **Tim Taubert**

- **Blake2 support for Mozilla's NSS library**: Implemented a Blake2 (a cryptographic hash function) module for Mozilla's Network Security Services (NSS) library. The library is used by Google, RedHat, and Mozilla products like Thunderbird.

2017 **Samsung R&D, India** **Intern**
Mentors: **Pratibha Moogi**, **Karthikeyan Somanathan**

- **Object localization framework**: Developed an end-to-end system to leverage deep learning models using custom convolutional neural networks for localization of objects in a given image.

Invited Talks

- Distributed Tracing without the Pain!
  KubeCon 2022, Detroit, USA
- Fast and Efficient Lookups via Data-Driven FIB Designs
  FIRA@SIGCOMM 2022, Amsterdam, Netherlands
- Data-Driven Network Path Simulation with iBox
  ACM SIGMETRICS 2022, Indian Institute of Technology Bombay (IIT-B), India
- Leveraging Service Meshes as a New Network Layer
  CSL Student Conference 2022, University of Illinois at Urbana-Champaign
- Leveraging Service Meshes as a New Network Layer
  HotNets 2021, University of Cambridge, Virtual Event
- iBox: Internet in a Box
  HotNets 2020, University of Chicago, Virtual Event
- iBox: Internet in a Box for Realistic Network Simulation
  Bandwidth Control Workshop 2019, Microsoft Research, Redmond

Honors

- Invited for demo at **Microsoft TechFest**, 2020.
- Finalist, **Pragyan CTF**, an international level security contest held by **NIT Trichy**, 2018.
- Among top 100 chosen by the **Indian Institute of Science (IISc)** for **CSA Undergraduate Summer School**, 2017.
- Among top 25 chosen by **Institute of Mathematical Sciences (IMSc)** for **Summer Research Program**, 2017.
- Among 45 chosen worldwide for **Mozilla Winter of Security**, 2016.

Skills

Languages: C++, C, Python, Bash
Tools: Kubernetes, eBPF, Protobuf, ns-2/3, gRPC